

CATEGORY SWITCHING AND SO-CALLED SO-CALLED PRONOUNS

David Pesetsky
M.I.T.

0. In this paper I will be discussing phrases like those of (1), consisting of a pronoun followed by a noun:

- (1)a. We linguists
- b. Us linguists
- c. You guys

In the first part of the paper I will consider two analyses of these phrases which have been proposed: one, by Postal (1966) (the determiner analysis), under which the pronoun is assumed to be functioning as a determiner or article on a noun head; and another, by Delorme and Dougherty (1972) (the appositive analysis), under which the noun is assumed to be an appositive modifier on a pronoun head. I will also briefly consider a third analysis, suggested to me by Guy Carden, under which the noun in these phrases is derived from a reduced restrictive relative clause.

While the appositive analysis explains quite elegantly the absence of singular NPs corresponding to (1), a fact for which I shall not offer an account, I shall argue against this analysis, showing that it cannot explain the identical singular/plural asymmetry in other languages and, in fact, makes incorrect predictions about the behavior of these phrases in English. I will suggest that an analysis like Postal's, under which the noun is the head and the pronoun a determiner, is correct. Arguments against the third analysis, deriving the nouns from underlying restrictive relative clauses, will fall out of general arguments in favor of the noun as head in these constructions.

Having established that the pronouns in these phrases are determiners, I will claim, in the concluding part of my paper, that we do not have to accept, as a consequence, Postal's proposal that all pronouns are thus underlying determiners. Indeed, if we assume, as seems more probable, that "bare" pronouns are normally heads of NPs, inserted under an N node, it is the more interesting question to consider how a single lexical item like a pronoun can be inserted into either DET or N position. I will propose that a lexical rule of category switching applying to pronouns accomplishes this. I will present some evidence from several languages which seems to suggest (tentatively) that languages which allow a process like category switching with other lexical items will have constructions like (1), while languages in which there is no other category switching will not have such constructions, at least not without some morphological change in the pronoun.

1. Determiner Analysis vs. Appositive Analysis

Under the determiner analysis, the pronoun in a phrase like we linguists is a determiner on the noun phrase linguists. Thus,

the phrases of (1) would be parallel in structure to phrases like the linguists or these linguists. Under the appositive analysis, the pronoun in the phrases of (1) is the head and the following noun is an appositive modifier. In this analysis, a phrase like we linguists is parallel in structure to phrases like we, the linguists or I, John Robinson. In this section I will present evidence showing that the first analysis is correct.

1.1 Singular/Plural Asymmetry: One of the strongest arguments proposed in favor of the appositive analysis, and a clear advantage it has over the analysis I will defend, is its explanation of the absence of singular constructions superficially corresponding to (1):

- (2)a. *I linguist
- b. *me linguist
- c. *you linguist

For Postal, these data must be handled as a "minor, more or less morphophonemic fact". In Delorme and Dougherty's appositive analysis, however, these facts fall out naturally for English. As appositive constructions, the phrases of (1) are in a class with phrases like we, the linguists. Delorme and Dougherty note that the absence of a comma break in (1) is not significant, since commas are often deletable in appositive constructions: we the people, I a linguist. Nouns in English normally take an obligatory article; thus a singular noun without an article is unacceptable in a sentence like *Linguist likes ice cream. It is an accidental fact about English, however, that the plural of the indefinite article is phonologically null, while no singular article is null. Thus, corresponding to a well-formed phrase we linguists, where the second element is assumed to be preceded by a null indefinite article, there would be no phrase *I linguist, since the noun in the singular must be preceded by a (phonologically real) article. The singular form of we linguists would be, rather, I, a linguist, with the indefinite article. Thus, Delorme and Dougherty's analysis allows the following paradigm:

- | | |
|--------------------|---------------------|
| (3)a. I a linguist | c. We Ø linguists |
| b. I the linguist | d. We the linguists |

but crucially not *I linguist.

It would be hard to argue with this explanation were it not the case that other languages whose NPs have different surface configurations show the same singular/plural asymmetry. Thus Russian, for example, lacks any articles (although it does have demonstrative determiners). Structures parallel to (1) occur, although they are awkward,² and show the same asymmetry as (1) and (2).

- (4)a. *ja leningradec 'I Leningrader'
- b. *ty leningradec 'you Leningrader'
- c. my leningradcy 'we Leningraders'
- d. vy leningradcy 'you Leningraders' ₃

French is another language which shows the same asymmetry, but will not allow Delorme and Dougherty's explanation for it. French does not normally allow a noun to appear without a phonologically realized article or other determiner preceding it. Thus, if the appositive analysis of constructions like (1) held in French, one would not expect to find a construction like (1) in other the singular or plural without some article preceding the noun. Yet we find the same paradigm as we found in English and in Russian:

- (5)a. *moi (autre) linguiste
- b. *toi (autre) linguiste
- c. nous autres linguistes
- d. vous autres linguistes

(I will discuss the *autres* element briefly below, but note for now that it is not plausibly an article or determiner, since it cannot begin an NP: *Autres linguistes sont...)

I do not have an explanation of this phenomenon. Nonetheless, its occurrence in languages with determiner properties as different as English, French and Russian strongly suggests that the explanation is not to be found in accidental phonetic properties of articles in English, but rather in some more general, probably semantic property of determiners and pronouns in general. Ken Hale (personal communication) has suggested to me that the restriction may be related to the obligatory unique reference of singular pronouns as opposed to the unspecified reference of plural pronouns. This may be the right approach. In any case, data from these other languages strongly suggest that the appositive analysis, elegant though it is, has not captured the right generalization about English.

The following arguments show that the behaviour of the pronouns in phrases like those of (1) patterns with determiners rather than with true pronoun heads of appositive constructions.

1.2 Number Agreement: An appositive modifying a plural head must be somehow semantically "numerous". A determiner such as a demonstrative, on the other hand, must show syntactic number agreement with its following head. Now note that a pronoun in combination with a noun, as in (1), acts like a determiner, and must show syntactic number agreement with a syntactically plural noun:

- (6)a. We, Parliament under God assembled, declare you are an honest yeoman. (semantic agreement)

- b. *Those Parliament under God assembled... (syntactic agreement)
- c. *We Parliament (*Us Parliament) under God assembled... (syntactic agreement)

1.3 Conjoined Determiners: The head of an appositive construction may be a conjoined structure. Conjoined determiners are odd in English. Conjoined pronouns in constructions like (1) are also odd, at about the same level, patterning with the determiners:

- (7)a. You and us (we), linguists from conviction, know we have found the truth.
- b. ?This and another linguist solved the problem.
- c. ?You and us linguists should get together some time.

1.4 All:₄ The quantifier all cannot immediately precede a personal pronoun. Thus, *All us read Panini is unacceptable. It can, of course, immediately precede a noun which is not a pronoun, and can precede a determiner modifying such a noun:

- (8)a. All men
- b. All the men

Now note that a pronoun head of an appositive construction is subject to the constraint described above, while a pronoun in a construction like that of (1) is not, and thus patterns with determiners. It is hard to see how an appositive analysis could explain this difference:

- (9)a. *All we (**us), the linguists of America, understand the riddle of existence.
- b. All us linguists understand the riddle of existence.

Under an appositive analysis both (9)a and (9)b have the same structure and similar sources, and thus should both be subject to the constraint. Under the determiner analysis, however, the pronoun in (9)b is not functioning as a pronoun at all, but as a determiner, and thus is not subject to the constraint, while (9)a, where the pronoun is the head of an appositive, is.

The following arguments suggest generally that the noun, and not the pronoun in constructions like (1) is the head of the NP in which it occurs.

1.5 Quantifier Scope: A quantifier like some, placed before an appositive construction, has scope over the head, but not the following appositive N. A quantifier placed before a phrase like we linguists has scope over the entire NP. Thus:

- (10)a. Some of us, (the) linguists = [some us][linguists]
 b. Some of us linguists = [some [us linguists]]

This can be seen in some of...others of... constructions, which serve, in their most normal usage, to identify mutually exclusive and complementary sets. On such a reading, the nouns modified by the quantifiers must be identical. Thus (11) is bad on the relevant reading:

- (11) *Some of the linguists think that others of the philosophers are crazy.

(12)a below is good, however, since some and others have scope over identical NPs--the heads of the appositive constructions--but do not have scope over their non-identical appositive modifiers. In (12)b, however, the quantifiers have scope over the entire pronoun-noun construction. Since the nouns are not identical, the sentence is bad on the relevant reading.

- (12)a. Some of us, linguists, think that others of us, philosophers, are crazy.
 b. *Some of us linguists think that others of us philosophers are crazy.

An appositive analysis of we linguists constructions could not account for the difference between (12)a and b, since they would presumably only differ by a minor phonological rule deleting pause between we and linguists. If we assume a determiner analysis, however, the facts fall out easily, without any extra assumptions that would not also be necessary under the appositive analysis. Either analysis must assume that determiners fall within the scope of these quantifiers to account for (13) (again on the relevant reading):

- (13) *Some of these linguists think that others of those philosophers are crazy.

Either analysis must also assume that the second element of an appositive is outside the scope of a quantifier preceding the whole NP, to allow (12)a. Under a determiner analysis, the badness of (12)b falls out of the first principle, while under the appositive analysis it must be considered an ad hoc exception to the second. Thus, these scope facts provide evidence for the determiner analysis, in which phrases like we linguists are noun headed, while an appositive like we, linguists is pronoun headed. (11) shows that the head is within the scope of quantification. (12)a shows that an appositive on a head is not. If linguists and philosophers were appositives in (12)b, one would expect the sentence to be good. Since it is not good, we conclude that linguists and philosophers are the heads of their phrases in (12)b.

1.6 Surface Constraint: A well-known surface constraint in English, which applies to the output of rules like Dative-Movement and Particle-Movement, is roughly statable as (14):

- (14) A pronoun direct object must immediately follow its verb.

This constraint blocks *He gave us it, derived by Dative-Movement from He gave it to us, and blocks (15)b as the output of Particle-Movement applied to (15)a:

- (15)a. He looked us up in the phone book.
b. *He looked up us in the phone book.

(16)a shows that this constraint applies to the heads of appositive constructions; (16)b-c, that it applies to pronoun heads of NPs in general. (16)d shows, however, that it does not apply to pronouns in constructions like those of (1):

- (16)a. *He looked up us, the local officers of the Elks.
b. *He looked up us, who were living in France then.
c. *He looked up us who sounded Kalmyk in the phone book.
d. He looked up us linguists in the phone book.

The appositive analysis has no way of explaining the difference between (16)a and (16)d. While there is no independent evidence that (14) should only apply to pronouns which are heads, only this reasonable assumption will yield the paradigm of (16). In fact, if we assume that us in (16)d is actually dominated by a DET or ART node, and not by an N node, then (14), looking at phrase structure, could not apply to (16)d, since us would not be identifiable as a pronoun.

1.7 Case Marking: Finally, without offering an explicit theory of English case marking, we note that pronoun-noun structures like (1) differ from appositive constructions in the optionality of subjective case marking. (17)a-b show that a pronoun head of an appositive is obligatorily given subjective case when in subject position. (17)c-d, on the other hand, show that, for the pronoun in a construction like those of (1), subjective case marking is optional in the same environment.

- (17)a. We, linguists from conviction, abhor computers.
b. *Us, linguists from conviction, abhor computers.
c. We linguists abhor a vacuum.
d. Us linguists abhor a vacuum.

Once again, an appositive analysis of (17)c-d cannot account for this difference in case marking, while the assumption of a structural difference between these sentences and (17)a-b could, given a theory of case marking in English. An appositive analysis predicts no difference between (17)b and d; the determiner analysis

provides for such a difference. Note that the difference does seem to be related to the status of the pronoun as head or determiner. Conjoined pronouns, where neither element is a clear head, show the same optionality of subjective case marking in colloquial speech (e.g. You and I/me like tomatoes.).

1.8 Reduced Restrictive Relative Analysis: A third possible analysis of phrases like (1), mentioned above, derives the nouns from restrictive relative clauses through a rule like *wh+is* deletion, with the pronouns as heads. Nouns in these constructions do seem to function more restrictively than appositively. Thus, while the head of an appositive can be repeated (with coreference) with a new appositive modifier adding new information, the same is not true of a restrictive relative or of a construction like (1):

- (18)a. We, linguists, think we, men of intelligence, should shut up occasionally.
- b. *People_i who are linguists think people_j who are intelligent should shut up occasionally.
- c. *Us linguists think us men of intelligence should shut up occasionally.

(18)d can probably be ruled out on semantic grounds, which there is not space to discuss here. For now, note that all the arguments suggesting that the noun and not the pronoun in the phrases of (1) is the head argue against this analysis too, as would any theory which ruled out rules like *wh+is* deletion as possible syntactic transformations. Moreover, such a rule for restrictives would require an ad hoc constraint only to operate with pronoun heads, since reduced relative clauses with noun heads are rare if not impossible in English.

2.0 Theoretical Implications

2.1 So-called "So-called Pronouns": Postal uses the fact that pronouns can function as determiners to propose the hypothesis that all "so-called pronouns" are derived as determiners on a *pro* head, which he identifies with the pronoun *one*.⁵ This is, however, by no means a necessary consequence of assuming that some pronouns can sometimes function as determiners. In fact, much of the evidence adduced here to support Postal's determiner analysis argues against Postal's conclusion, since pronouns functioning as heads of NPs have been shown to act very differently from pronouns as determiners.

If, then, all pronouns are not to be derived from underlying determiners, how do they come to function that way? Most theories of the lexicon, beginning with Aspects, have assumed that a lexical item's categorial specification is part of its lexical entry. One would like to consider *we* as a pronoun and *we* as a determiner as one lexical entry, since they share the same referential properties and irregular morphology. If there is only one lexical entry *we* and one entry *you* (at least in the plural), then what mechanism must one assume to allow them to be inserted into phrase struc-

tures both as nouns (under an N node) and as determiners (under a DET or ART node)?

My proposal is that these pronouns be entered in the lexicon as N, and that a lexical rule of category switching, looking something like (19), applies to their entries within the lexicon:

- (19) $[N_{+pro}] \rightarrow [DET]$

I assume that limitation of (19)'s application to plural pronouns will result from independent semantic considerations, as I suggested in section 1.1. Presumably the rule will be restricted in other ways--for example, to rule out its application to who: *who linguists. (But note the use of what as an N in What did you see and as some sort of determiner in What linguists did you see.)

Other phenomena which could be considered examples of category switching are well-known in English, where a noun may be used without derivational morphology as, for example, a verb in (20)a or as some sort of modifier in (20)b:

- (20)a. Carson hosted the TV show. ($[N] \rightarrow [V]$)
 b. Carson suffers from host neurosis. ($[N] \rightarrow [Adj] (?)$)

If we assume that sentences like (20)a-b are instances of category switching, we can make a prediction. Many languages, such as the Romance languages, completely lack this ability to use single lexical items as members of more than one syntactic category. One would expect these languages to be likely to exclude the construction found in (1), where a pronoun acts as a determiner without any derivational morphology. Languages which show other instances of category switching, on the other hand, would be more likely to have a construction analogous to (1).

Examination of a number of languages seems to be consistent with this prediction. Germanic languages tend to allow some degree of category switching. Norwegian apparently is nearly as free as English. German at least allows lexical nouns to function as modifiers, as in what are written as compound nouns.⁶ German and Norwegian both have constructions analogous to (1). In German, at least, it is clear that the pronoun in the construction in question is functioning as a determiner. Adjectives in German take "weak" endings (nom. pl. -en) if preceded by an article or other determiner, and take "strong" endings (nom. pl. -e) otherwise. Adjectives after a pronoun in a construction like (1) take weak endings, indicating that the pronoun is functioning like a determiner. (21) shows the relevant contrasts:

- (21)a. Gute Männer... 'good men' (strong)
 b. Die guten Männer... 'the good men' (weak)
 c. Wir guten (*gute) Männer... 'us good men' (weak)

Latvian appears to work like German (although I have no direct evidence that the pronoun in its analog of we linguists is a de-

terminer, aside from the absence of a comma intonation). It allows nouns to function as modifiers and also allows a construction parallel to (1). Tok Pisin, an English-based language of Papua New Guinea, allows extremely free category switching, including a construction analogous to (1).⁸ Japanese apparently is similar to German, allowing nouns to function as modifiers and allowing a construction like (1).

Romance languages, however, present a different pattern of data. Neither French, Italian or Spanish has category-switching phenomena parallel to (20), and none of these languages has a construction parallel to (1), where a bare pronoun precedes a noun. Spanish provides the simplest example. A construction corresponding to (1) is impossible in Spanish. Thus Spanish does not allow *Nosotros linguistas. (A meaning similar to the English can be conveyed by an appositive--nosotros los linguistas--or by the device of inflecting the verb in the first or second person plural with a third person subject like los linguistas.)

The situation in French and Italian is somewhat more complex. In neither of these languages is a construction parallel to (1) possible (*nous linguistes, *noi linguisti).⁹ Both of these languages must use "derivational morphology" in some sense to allow their plural pronouns to function as determiners. As we saw in (5) above, French nous as a determiner appears as nous autres. Although this etymologically means "we others", the autres in this construction has lost any semantic function. Italian uses the analogous form noi altri as a determiner: noi altri linguisti. Both languages have parallel second person forms: vous autres, voi altri. These facts are consistent with a hypothesis that category switching is involved in the formation of English phrases like we linguists. Romance languages lack a mechanism of category switching in general; so in order for a pronoun to function as a determiner some change must be made to it, creating a new form.

One final note about French. In some French Canadian dialects nous autres and vous autres are replacing nous and vous as emphatic forms of the pronouns. Accompanying this has been a decline in the acceptability of nous autres and vous autres as determiner forms of the pronouns. An explanation for this is suggested by the category-switching hypothesis. A language without category switching will tend to create separate forms for noun and determiner uses of its pronouns. If the determiner form is adopted for nominal use, one result might be to lessen the acceptability of the determiner usage. It may be that Canadian French is making the same move Spanish may have made in the fourteenth century. It is unclear what function nosotros (etymologically the same as nous autres) had before it replaced nos as a normal pronoun, but it is interesting that, with nosotros as the usual pronoun, it cannot have the determiner usage.¹⁰

Finally, Russian is a problematical case for our suggested generalization. Russian does not offer clear cases of category switching. Noun-noun compounds do occur, but with the first noun in a combining form (e.g. kamne-drobilka 'stone crusher', where kamne is a combining form of kamen). A category-switching explanation

of these is dubious. (Constructions like gorod-geroj 'city-hero' lack a clear head, both elements being declined, and seem to be merely some sort of juxtaposition.) In Russian, contrary to our prediction, constructions like (1) are possible, as we saw in (4), but are considered extremely awkward, limited to declamatory or slogan contexts. Thus, while the category switching theory predicts their relative unacceptability, it cannot account for the acceptability they do have.

3.0 Conclusion

I have shown that pronouns in English can also function as determiners and have suggested that a mechanism of category switching is responsible for this phenomenon. I have also suggested that the varying acceptability of pronouns as determiners in various languages may be related to the varying possibilities of category switching as a general process in these languages.

Note that if category switching is a lexical process in some languages, it is not an unconstrained process. Thus it can turn nouns into verbs in English and Norwegian, but not in German. One would presume that certain possibilities, such as a rule changing verbs into particles or pronouns into prepositions, do not occur. Study of which switches are possible and which are not could shed light on the nature of syntactic categories in general. 11

Footnotes

I am grateful to Noam Chomsky, Ken Hale and Jim Harris for useful discussion of an earlier version of this paper, and to Guy Carden, Donca Steriade and Tim Stowell for much discussion then and now. None of them necessarily agrees with me, and all mistakes are my own. Also thanks to foreign language informants who are, unfortunately, too numerous to list here.

1. Delorme and Dougherty argue that these appositives are base-generated. My arguments against the appositive analysis are neutral between this assumption and an assumption that they derive from reduced appositive relative clauses.

2. Their awkwardness is discussed below. Some informants do not like them at all, but live examples occur: My leningradcy gorjačo odobrijaem projekt novoj konstitucii SSSR 'We Leningraders warmly approve the draft of the new constitution of the USSR'--recent slogan.

3. I omit the third person plural in all my examples. This universally seems unacceptable or marginal for reasons which are totally obscure to me (and are not handled by any analysis). Note it is acceptable in some English dialects as a synonym of the determiner those: them guys. It is interesting that it can only occur in the objective case with this usage, although the OED has some citations of phrases like they boys.

4. This argument and 1.6 were suggested to me by Guy Carden.
5. Strictly speaking, he assumes that pronouns are underlyingly determiner or article features on the head noun, which are spelled out as surface determiners by a segmentalization rule, but this portion of his argument is not relevant here.
6. This analysis of German compounds may be controversial, but I can see little difference between German compounds orthographically written as a single word, like Fullfederhalter and English constructions like pen holder, where the first noun is clearly in a modifying relation to its head.
7. I am grateful to Tarald Taraldson for suggesting this line of inquiry.
8. Thanks to Ellen Woolford for Tok Pisin data.
9. One French informant reports finding nous linguistes more natural in her dialect than nous autres linguistes, a counterexample to my analysis. Other informants gave the reverse judgement, however.
10. One informant, from Roberval, Que., can use nous autres and vous autres in subject position in preference to standard French nous (nous autres alongside on), thus: Nous autres aimons beaucoup le hockey 'We like hockey'. For her, nous autres and nous are both unacceptable as determiners. We have, however, been unable to find other speakers of this dialect, which would provide nice suggestive evidence for the category-switching hypothesis.
11. Since this paper was written, Jackendoff's (1977) brief discussion of this construction has come to my attention. He accepts the determiner hypothesis and argues that it provides evidence for his feature decomposition of syntactic categories. For him, nouns and articles differ only in the feature $\pm$ COMP. (Nouns are +COMP.) He suggests that the phenomena described here can be accounted for by leaving pronouns unspecified for COMP, thus allowing them to be either nouns or articles. (He distinguishes between articles and other determiners.) If this is right, then our category-switching generalization could be captured by assuming that languages impose different restrictions on which category features, if any can be unspecified in lexical entries. My intuition, however, is that the noun usage of pronouns is somehow more basic and unmarked than the determiner usage. (Note that languages like Spanish have pronouns which function as nouns but none which function as determiners. I doubt that any language has the reverse situation, though it is possible.) A category switching theory could capture this markedness distinction, since a pronoun's noun specification can be considered fundamental, while its determiner specification is derived by rule.

Bibliography

Corominas, J. (1954), Diccionario Critico Etimologico de la Lengua Castellana, Berne.

Delorme, E. and Dougherty, R. (1972), "Appositive NP Constructions", Foundations of Language 8:2-29.

Jackendoff, R. (1977), X Syntax: A Study of Phrase Structure, Cambridge.

Postal, P. (1966), "On So-Called 'Pronouns' in English" in Reibel, D. and Schane, S.A. (eds.) (1969), Modern Studies in English, New Jersey.